

Decoupling AI: When to Move from In-Process to Event-Driven Pipelines

Monolithic AI pipelines hit a wall when orchestrating hundreds of external tasks. Discover the tipping point for migrating to Pub/Sub.

Executive Summary:

Startups often build AI data ingestion as a linear, in-process script. As the system scales to handle hundreds of concurrent web crawls, API fetches, and LLM extractions, this monolithic approach results in cascading transaction failures and stalled HTTP requests. This whitepaper charts the architectural evolution from a fragile synchronous loop to a resilient, asynchronous Pub/Sub event-driven architecture.

1. The Engineering Challenge

Our initial scouting pipeline processed 70 company endpoints via a single Cloud Run loop. If one company's ATS adapter failed or hit an extraction timeout, the entire Postgres transaction entered an aborted state. The final `session.commit()` rolled back everything, meaning 69 successful fetches were discarded due to one failure. Additionally, the process routinely exceeded the 30-minute HTTP timeout of Cloud Scheduler.

In high-stakes enterprise environments, this kind of failure is unacceptable. When building systems for Fortune 500 organizations, relying on basic prompt engineering or out-of-the-box vendor SDKs frequently masks underlying architectural fragility. The goal is to move beyond simple proofs-of-concept into deterministic, resilient data flow.

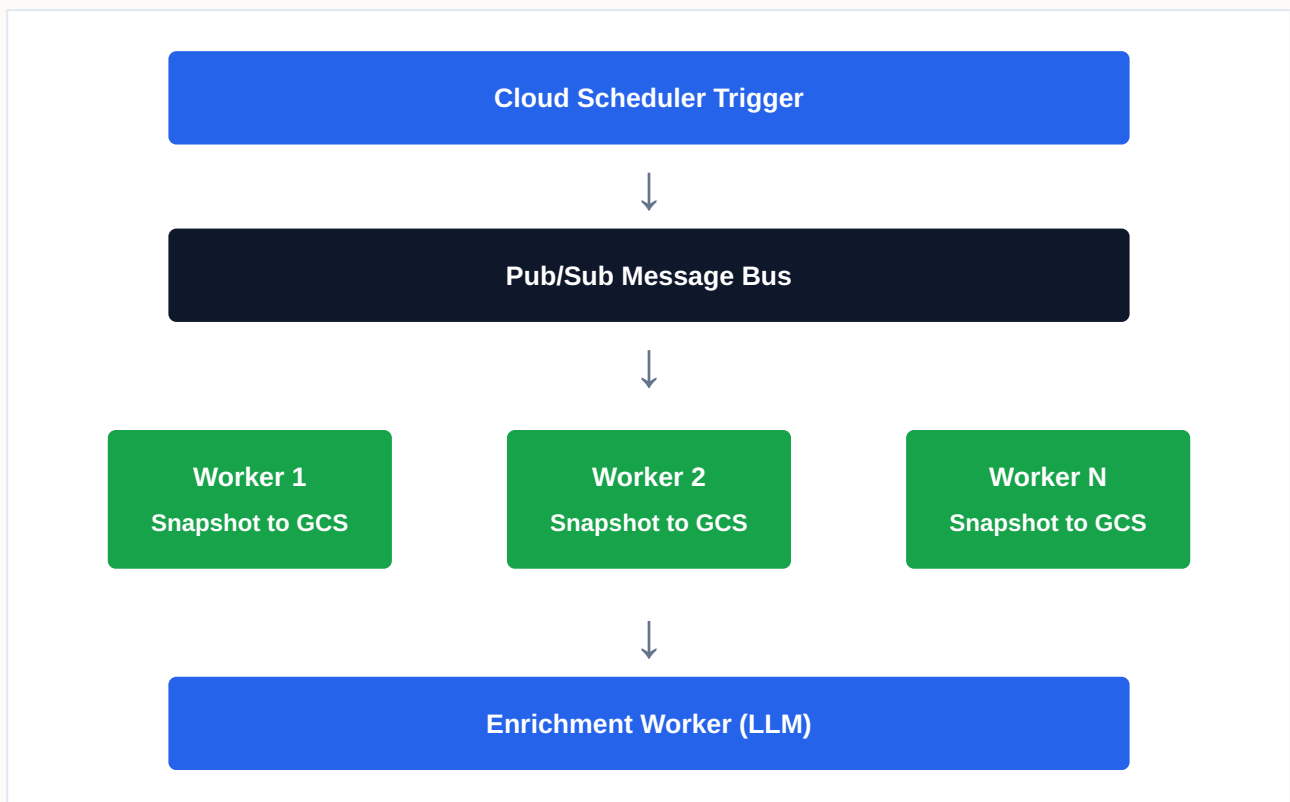
2. The Architectural Solution

We transitioned the architecture from a sequential script to a Pub/Sub event chain. The orchestrator now simply emits `scout_task` messages. Independent, stateless workers consume these messages, executing the fetch and dumping raw, immutable JSON snapshots into Cloud Storage. A separate `source_hit` event triggers the LLM enrichment phase. This

isolation guarantees that failures are highly localized, transactions are incredibly brief, and the entire pipeline is idempotent and replayable from the raw storage bucket.

By enforcing this pattern at the architectural level, the system no longer relies on the "magic" of generative fluency. Instead, it relies on rigorous data engineering and precise, targeted models. This decoupling ensures that failures are isolated, auditable, and easily corrected without unwinding the entire data pipeline.

3. Structural Data Flow



4. Business Impact & ROI

For organizations operating at scale, migrating to this pattern fundamentally alters the cost and reliability curve. It shifts the burden of trust from the LLM's inherently probabilistic nature to a deterministic data engineering layer. The result is a system that can process massive throughput securely, drastically reducing API token burn and ensuring complete auditability for internal compliance boards.